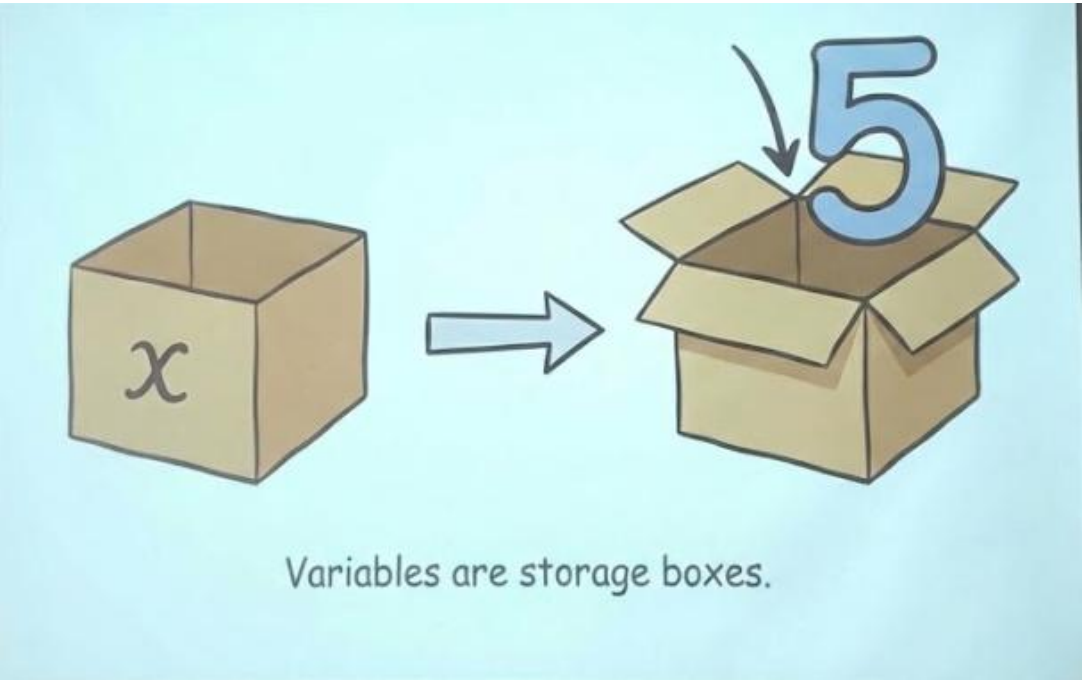
CENG 113M: Programming Basics

- Lab 1 - The Absolute Basics & Workspace Setup

- Welcome to the world of coding!
- Today's Goal: Setting up our digital workspace.
- The Secret: Coding is just mathematical logic translated for machines.

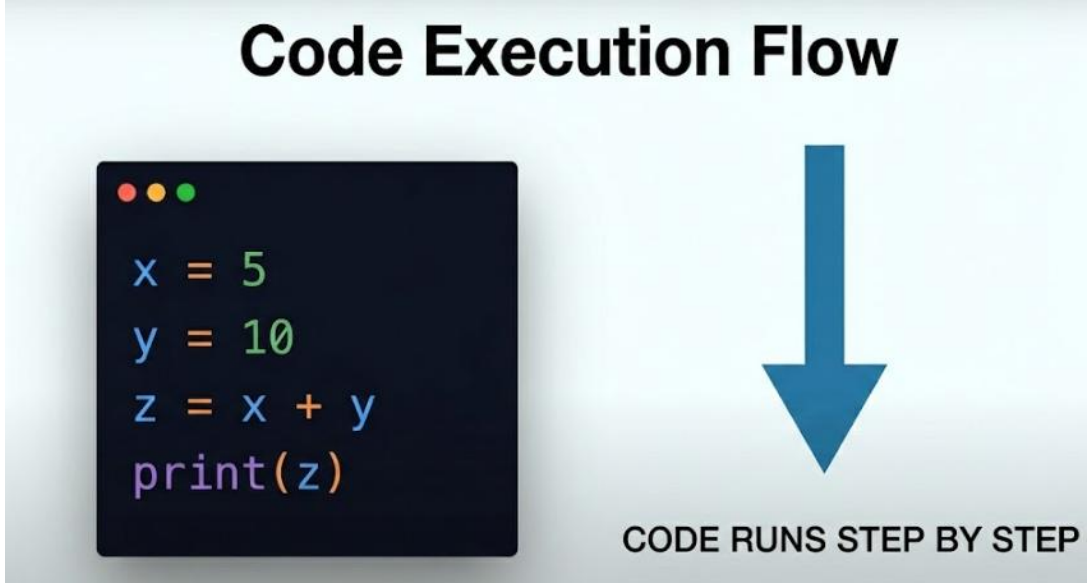
What is a Computer Program?

- A program is simply a recipe (a sequence of instructions).
- Computers are not smart; they are just fast and obedient.
- The Golden Rule of "=":
  - Math: = means equality (Truth).
    Example: x + 2 = y
  - Code: = means assignment (Action).
    Example: x = 5
- Read code from right to left: "Calculate the right side, put it into the box on the left."

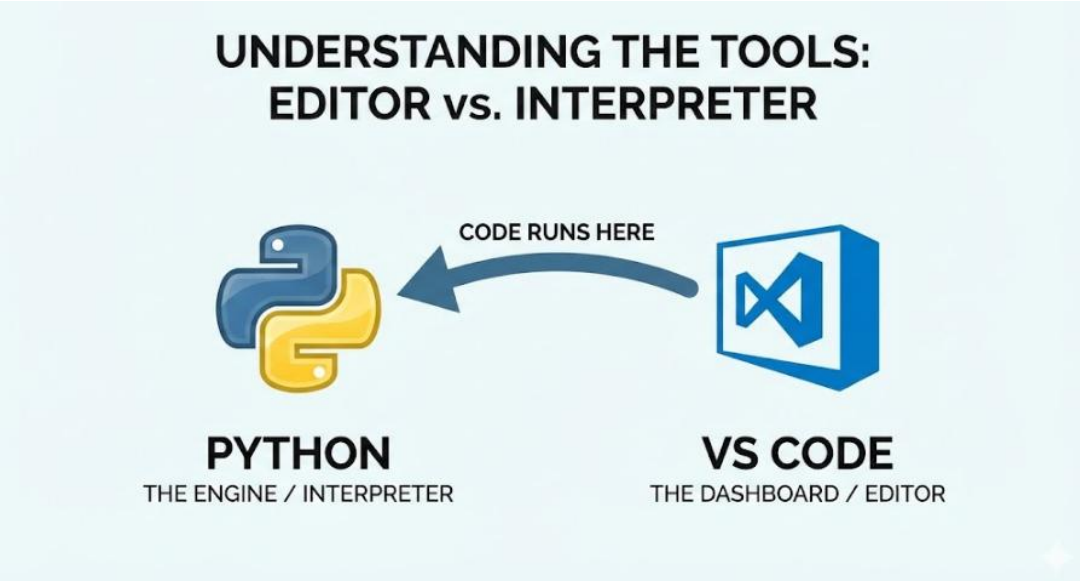


How Computers Read (Top-to-Bottom Rule)

- A program is read strictly from top to bottom.
- The computer only knows what has happened so far.
- It cannot see the future lines.If you try to use a box (variable) before creating it, the computer will crash!
- Code Literacy = Reading the code line-by-line, exactly like the computer does.



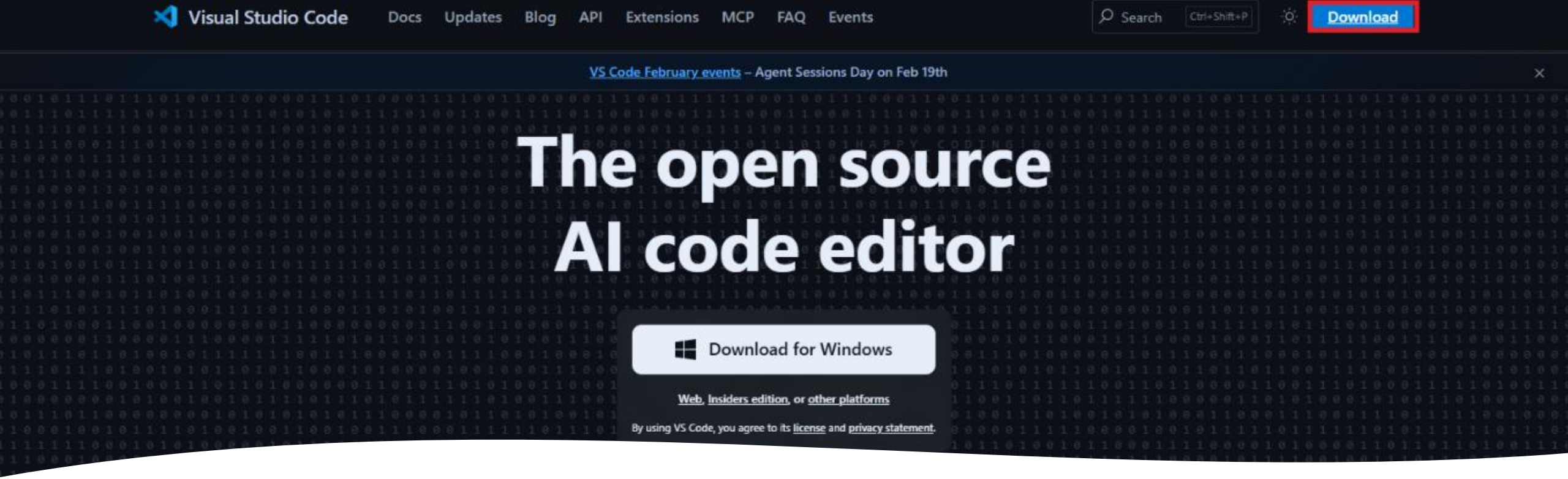
Our Toolkit: What Do We Need to Code?



- We need two pieces of software to talk to the computer.
    - 1. The Interpreter (Python): This is the engine. It translates our human-readable Python code into 1s and 0s (machine code) that the computer understands.
    - 2. The Code Editor (VS Code): This is our dashboard. It’s a smart notebook where we type, save, and organize our code before sending it to the engine.
    - You can't drive a car with just a steering wheel (Editor); you need the engine under the hood (Python).

Step 1: Installing Python

- Getting the engine ready.
  - Open your browser and go to: python.org/downloads
  - Click the big yellow download button for the latest version.
  - Open the downloaded file.
    - CRITICAL STEP for Windows Users: Look at the bottom of the installation window.
    - You MUST check the box that says: "Add python.exe to PATH«
    - If you don't check this, your computer won't know where the engine is!
  - Click "Install Now" and wait for it to finish.
  - (Mac Users and Linux Users should get python pre-installed).

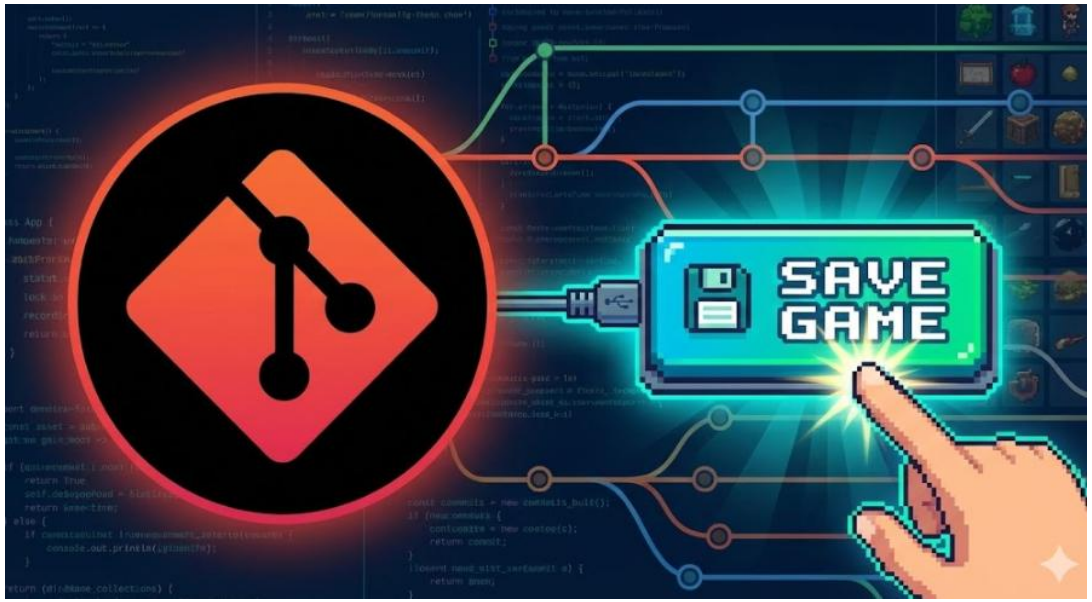


Step 2: Visual Studio Code

- Getting our dashboard ready.
  - Open a new tab and go to: **code.visualstudio.com**
  - Click the big blue button: **Download for Windows / Mac**
  - Run the installer.
  - You can just click "Next" through the standard options.
  - Accept the agreement and install.
  - Once installed, go ahead and launch it! You should see a "Welcome" screen.

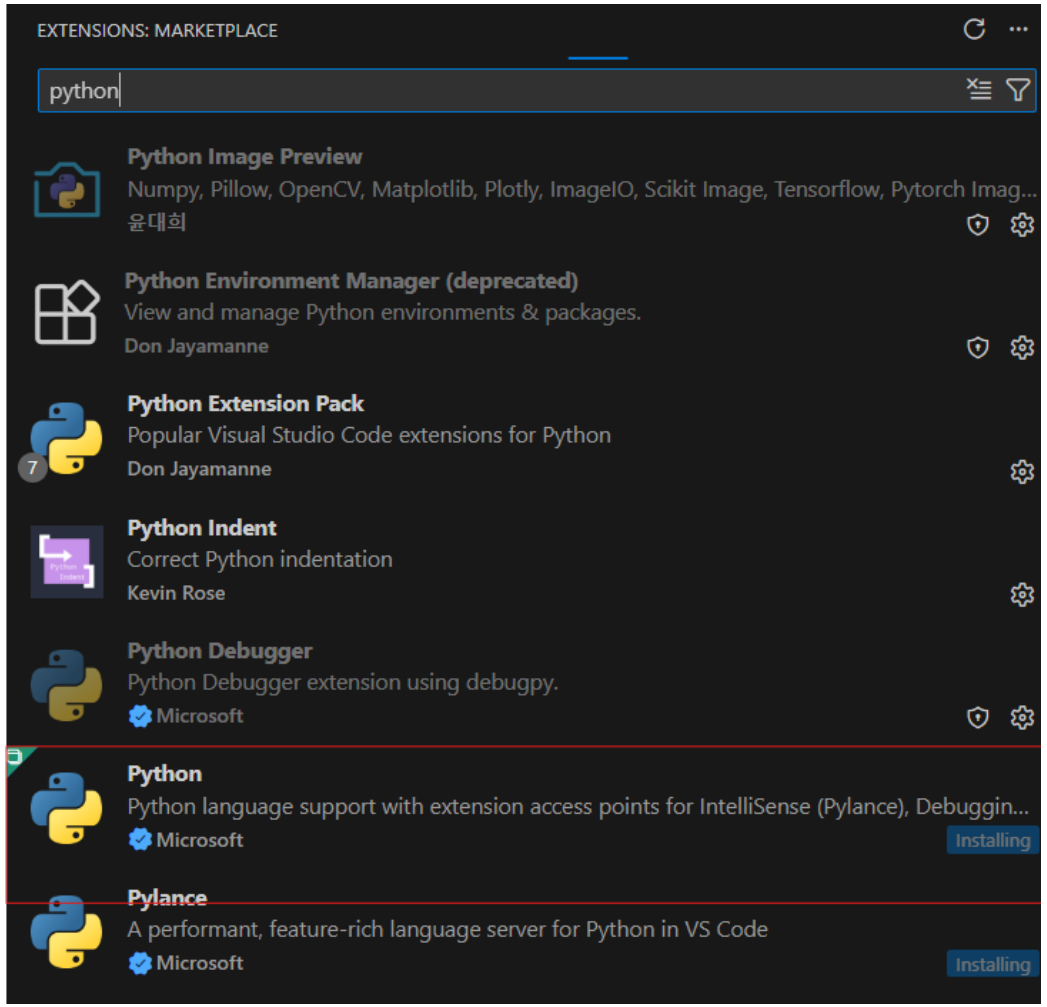
Step 2.5 - Installing "The Time Machine" (Git)

- What is it?
  - It saves "versions" of your code. If you break something, you can go back in time!
- Action:
  - Go to: git-scm.com
  - Download and Install.
  - Settings: Just keep clicking "Next" (Default settings are fine).



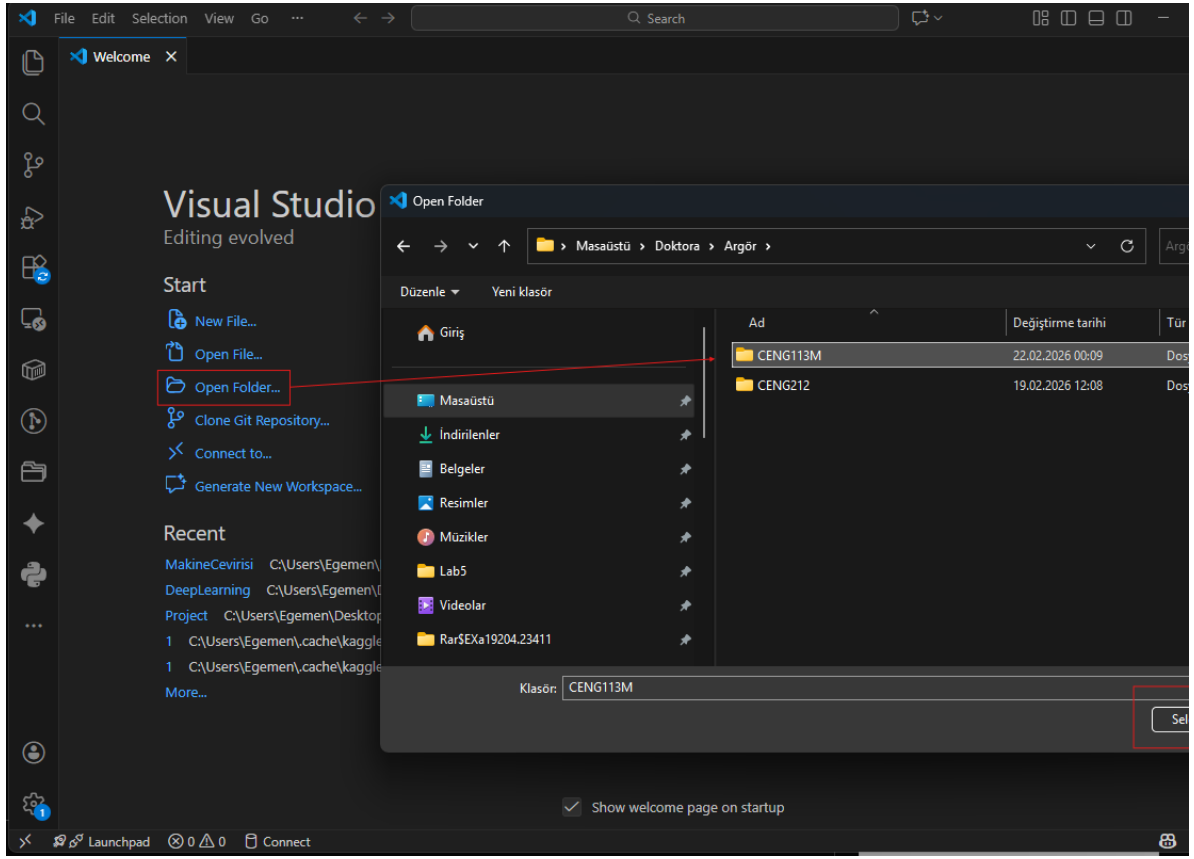
Step 3: The Python Extension

- Making our dashboard smart.
  Out of the box, VS Code is just a blank notebook.
  We need to teach it Python.
  - Look at the left menu (Activity Bar) in VS Code.
  - Click the Extensions icon (it looks like 4 building blocks).
  - Type Python in the search bar.
  - Find the one published by Microsoft (it has a blue checkmark and millions of downloads).
  - Click Install.
  - Wait a few seconds… you are now ready to code!



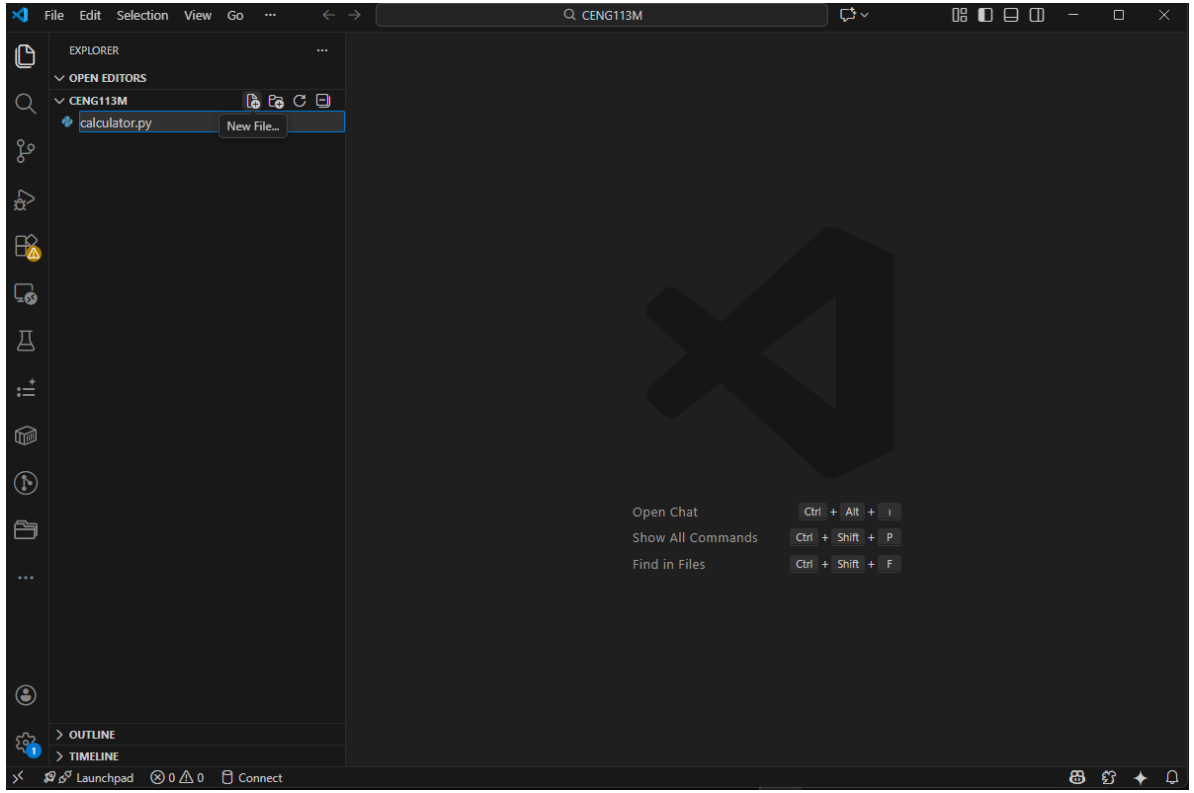
Step 4: Creating Your Workspace

- **Rule #1 of Coding:** Keep your projects organized!
- **Action 1:** Go to your computer's Desktop.
- **Action 2:** Right-click > New Folder.
- **Action 3:** Name it **CENG113_Lab1**
  - *Pro-Tip:* Programmers hate spaces and Turkish characters (ç, ş, ğ, vb.) in file names. Use underscores _ instead!
- **Action 4:** Open VS Code. Go to **File > Open Folder...** and select your new folder.



Step 5: Your First Python File

- Look at the left panel (Explorer) in VS Code.
- You should see your folder name: CENG113_Lab1
- Click the **New File** icon (+) next to it.
- Type the name: **calculator.py** and press Enter.
- **Why .py?**
  - The extension .py is the magic word.
  - It tells VS Code: *"Hey, this is not a normal text file. Wake up the Python engine!"*
  - You will see the file icon turn into the Python logo.

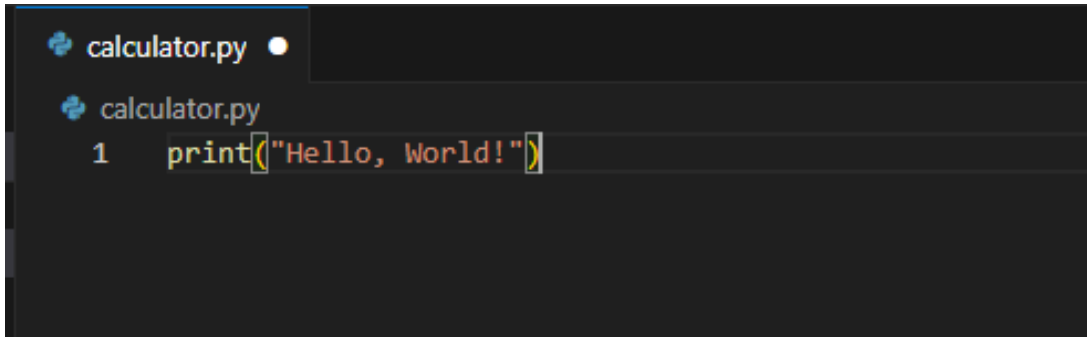


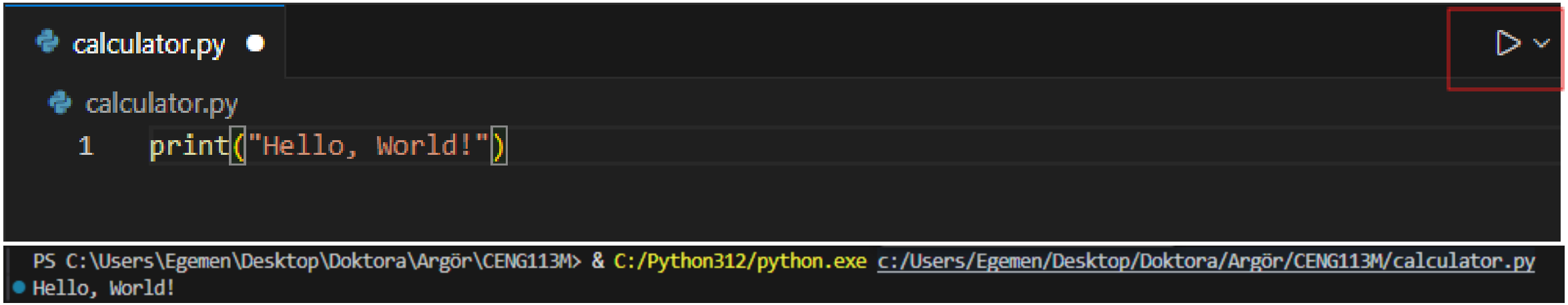
Step 6: “Hello, World!”

- Click inside the big blank editor window for calculator.py.
- Type exactly this on Line 1:

print("Hello, World!")

- **Breaking down the code:**
  - print : The command (function). It means "show this on the screen" (not your physical printer!).
  - () : Parentheses hold the *ingredients* you give to the command.
  - "" OR '': Quotes tell the computer "This is text".
- **CRITICAL RULES FOR QUOTES:**
  - Consistency: If you start with ", you must end with ".
  - Bad: print('Hello') Syntax Error!
  - The Keyboard Trap: The double quote (") is a SINGLE KEY.
  - Do NOT press the single quote key (') twice!

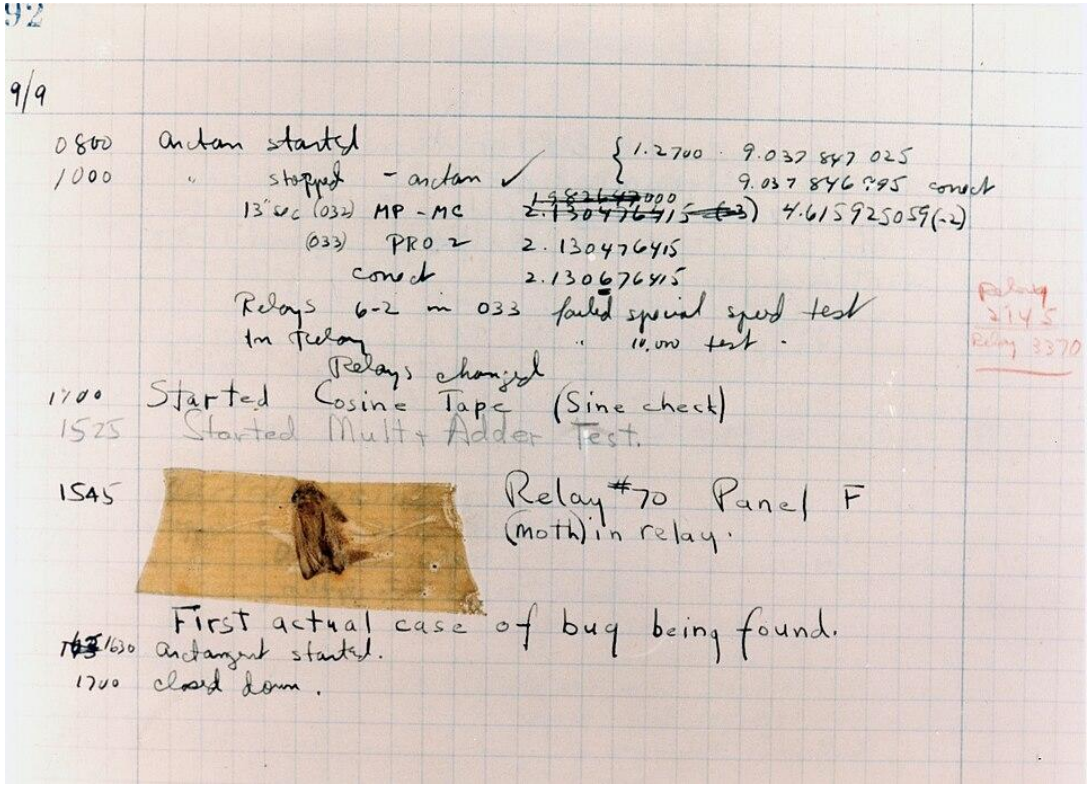




Step 7: Save It, Run It, See It!

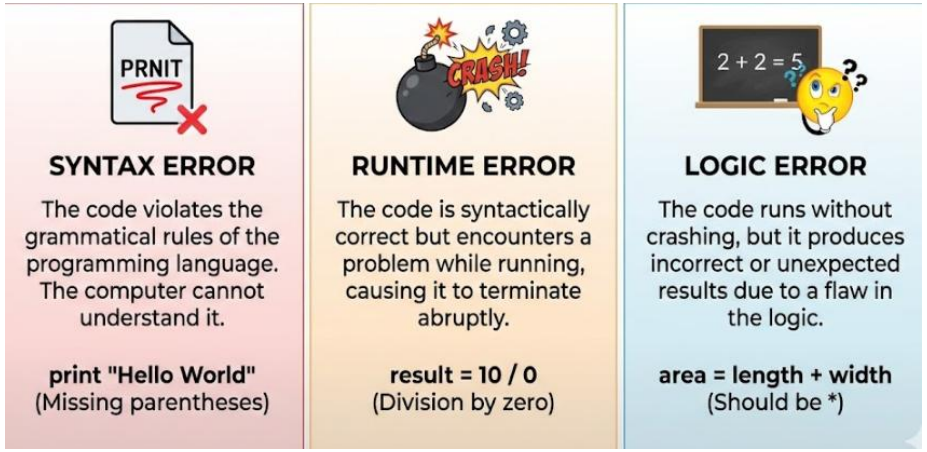
- The most important keyboard shortcut you will learn.
  - 1. Look at the tab: Do you see a white dot next to calculator.py? That means your code is NOT saved.
  - 2. The Golden Reflex: Press Ctrl + S (Windows and Linux) or Cmd + S (Mac). The dot disappears!
  - 3. Run the Engine: Click the Play button (▷) in the top right corner of VS Code.
  - 4. Check the Terminal: Look at the bottom of your screen. The computer just talked back to you!

What is a Bug?



- A "**Bug**" is simply a mistake or an error in your code.
- "**Debugging**" is the detective work of finding and fixing that mistake.
- **Fun Fact**: The term became famous in 1947 when an actual moth got stuck inside a giant Harvard computer and caused it to crash!
- **The Mindset**: * In math, if you drop a minus sign, the proof fails.
  - In coding, if you drop a parenthesis, the program crashes.
  - It is 100% normal. We will spend more time debugging than writing!

Know Your Enemy: The 3 Types of Bugs



1. Syntax Errors (The Grammar Mistake):

•You misspelled a word or forgot a parenthesis).

•The computer says: *"I don't understand this language!"* It won't even start running.

2. Runtime Errors (The Impossible Task):

•The grammar is correct, and the code starts running. But it hits an impossible command (like dividing a by zero).

•The computer says: *"I tried, but I crashed!"*

3. Logic Errors (The Silent Killer):

•No crashes, no red text. The code runs perfectly... but gives the **wrong answer**.

•E.g., You used a + instead of a -

•The computer says: *"I did exactly what you asked!"* (But you asked for the wrong thing).

Reading a Crash Report (Traceback)

- When your code crashes, Python prints a Traceback.
- The Golden Rule: Always read the error message from the BOTTOM UP.
- Be a detective and look for 3 clues:
  1. The Error Type (Bottom line): What went wrong? (e.g., SyntaxError)
  2. The File Name: Which file has the bug?
  3. The Line Number: Exactly where did the computer get confused?

```
PS C:\Users\Egemen\Desktop\Doktora\Argör\CENG113M> & C:/Python312/python.exe c:/Users/Egemen/Desktop/Doktora/Argör/CENG113M/calculator.py
Traceback (most recent call last):
  File "c:\Users\Egemen\Desktop\Doktora\Argör\CENG113M\calculator.py", line 1, in <module>
    prnt("Hello, World!")
    ^^^^
NameError: name 'prnt' is not defined. Did you mean: 'print'?
```

Live Detective Work 1

- Goal: Calculate the area of a circle ($A = \pi r^2$).
- Task: Delete your old code in calculator.py and type exactly what you see beside. Run it and read the red text!

```
pi_value = 3.14
radius = 5

Area = pi_value * (radius * radius

print(area)
```

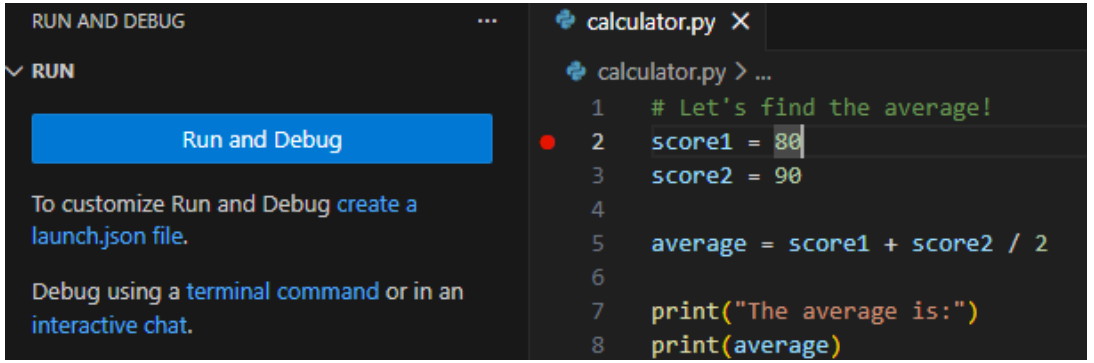
Live Detective Work 2: The Logic Error

- The code runs perfectly… but lies to you!
- Goal: Calculate the average of two exam scores (80 and 90). The mathematical formula is $\frac{a+b}{2}$
- Task: Type this code, save it, and run it.

```
# Let's find the average!
score1 = 80
score2 = 90

average = score1 + score2 / 2

print("The average is:")
print(average)
```

The VS Code Debugger

- Watching the computer think, line by line.
  - Step 1: The Breakpoint. Hover to the left of line number 2 and click. A red dot 🔴 will appear. This is a stop sign for the computer!
  - Step 2: Start Debugging. Look at the left menu. Click the icon with the Bug and the Play button. Click "Run and Debug" (choose Python File).
  - Step 3: Step Over (F10). The code will freeze at the red dot. Look at the "Variables" panel on the left.
  - Click the Step Over ⤵️ button (or press F10) to execute one line at a time. Watch the boxes (variables) get filled!



Live Detective Work 3: The Runtime Crash

- Perfect grammar, impossible math!
- Goal: See what happens when the engine hits a wall.
- Task: Type this code, save, and run.

```python
# The impossible task
total_apples = 50
number_of_students = 0

apples_per_student = total_apples / number_of_students

print(apples_per_student)
```

Math Rules Apply Here: PEMDAS

- How Python calculates formulas.
  - Python is essentially a giant, super-fast calculator. It strictly follows standard mathematical rules.
  - P - Parentheses () Highest priority! Always calculated first.
  - E - Exponents ** (Notice it's two stars in Python, not ^)
  - M/D - Multiplication * & Division /
  - A/S - Addition + & Subtraction –
  - Golden Rule: When in doubt, just use parentheses! (a + b) / 2

Part 5: The Mini Bug Hunt

- Goal: Calculate the slope (m) of a line passing through (x_1, y_1) and (x_2, y_2)
- Mathematical Formula: $m = \frac{(x_1, y_1)}{(x_2, y_2)}$
- Task:
- Create a new file named slope.py.
- Type the exact code beside.
- Find and fix the Syntax Error, the Runtime Error, and the Logic Error.
- Raise your hand when you get the correct answer!

```
# Calculate the slope of a line!
x1 = 4
y1 = 8
x2 = 4
y2 = 16

slope = y2 - y1 / x2 - x1

print("The slope is: , slope)
```

The Solutions!

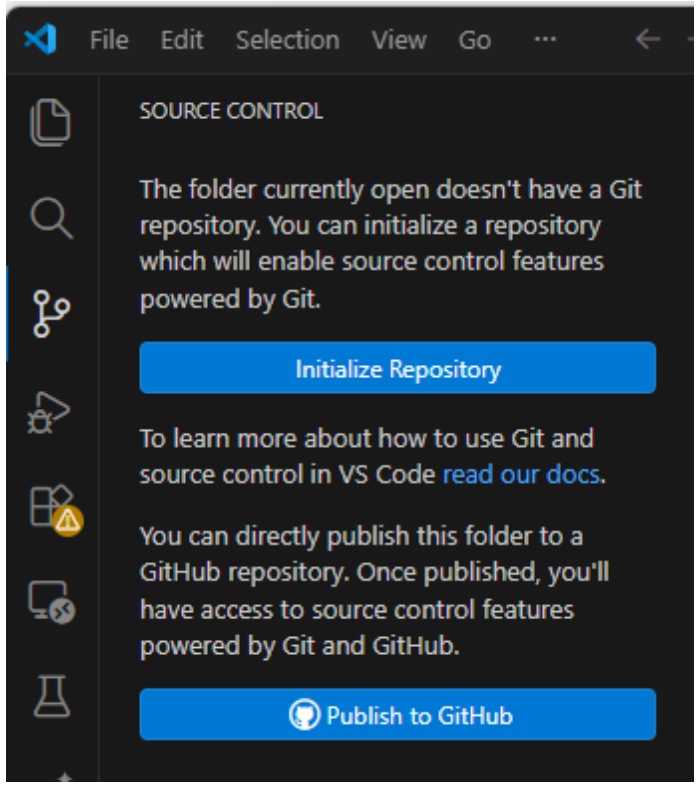
- How to fix the broken math.
  - The Fixes:
  - Syntax Error: Missing closing quotation mark " on the print statement.
  - Runtime Error: x_2 - x_1 is 4 - 4 = 0. Division by zero! We must change x_2 to a different number, like 6.
  - Logic Error: Missing parentheses around the numerator and denominator.

```python
# THE FIXED CODE
x1 = 4
y1 = 8
x2 = 6  # FIXED: Changed from 4 to prevent ZeroDivisionError
y2 = 16

# FIXED: Added parentheses for PEMDAS!
slope = (y2 - y1) / (x2 - x1)

# FIXED: Closed the quotation marks!
print("The slope is: ", slope)
```

Bonus Level: Publish Your Code

Sharing your work with the world.

- Prerequisite: You need a GitHub.com account (Free).
- The Magic Button in VS Code:
  - Click the Source Control icon on the left (looks like a branch).
  - Click "Publish to GitHub".
  - VS Code will ask to sign in. Allow it.
  - Select "Publish to public repository".
- Result: Your code is now on the internet! You can send the link to your friends.



Code Literacy – Questions 1

- Fact: print(a + b) displays the sum of two variables.
- Fact: print(2 * w) displays 2 times the given variable.
- Question: Which code correctly calculates the Perimeter of a Rectangle
    - $Formula: 2 \times (width + height)$
- Options:
    1. print(2 * width + height)
    2. print(2 * (width + height))
    3. print(width + height * 2)

Code Literacy – Questions 2

- Fact: random.randrange(1, 6) picks a number from 1, 2, 3, 4, 5. (6 is excluded!)
- Question: We want to simulate a Coin Toss (0 for Heads, 1 for Tails). Which code is correct?
- Options:
  1. random.randrange(0, 1)
  2. random.randrange(0, 2)
  3. random.randrange(1, 2)

Code Literacy – Questions 3

- Fact: numbers[i] brings the $(i+1)^{th}$ element.
- Fact: In Computer Science, we start counting from 0.
- Question: We have a list: days = ["Mon", "Tue", "Wed"]. Which code prints "Tue"?
- Options:
  1. print(days[1])
  2. print (days[2])
  3. print(days[0])

Code Literacy – Questions 4

- Fact: If inputs are Numbers, + adds them (Math).
- Fact: If inputs are Strings ("text"), + glues them together (Concatenation).
- Fact: A comma , outside quotes separates items and puts a space between them.
- Fact: Any symbol inside quotes ("...") is just a character (Ink on paper).
- Question: Based on these facts, what is the output of these 5 lines?

#	Code	Your Prediction
1	print(3 + 2)	?
2	print("3" + "2")	?
3	print("3 + 2")	?
4	print("A", "B")	?
5	print("A, B")	?